

Name of the subject: Analysis of Structures			
Department: Architecture and Regional Planning			
Curriculum: UG (2nd Year)	Type of Course: Compulsory	Semester: 3rd (AUTUMN),	
LTP: 3-1-0,	Credits: 4	Prerequisite(s): Nil	

Course Overview:

The Structural Analysis course intends to acquaint architecture students with basic theory and behaviour of structural system. Some of the features of strength of materials are also introduced in the course. The course will give major emphasis to the analysis of indeterminate beams and frames. It is also designed to address the architectural application of structural system like arch, shell, dome, plate and membrane structure. The lecture will demonstrate the principles of structural analysis in coordination with its simple application in architecture.

Learning Objectives:

After the completion of this course, students will be able to:

1. Analysis of determinate truss.
2. Computation of bending and shear stress in the beam.
3. Analysis of indeterminate structural members.
4. Schematic development of architectural structural systems

Course Contents:

Bending and Shear Stresses in Beams: General theory, bending equation and distribution of bending and shear stress in beam cross section; Deflection of Beams, Theory of Column: Euler's theory and Rankin's Formula, Analysis of Indeterminate Structures: Introduction to force and displacement method; Special Structures: Structural concept and architectural application of Arch, Shell, Dome Tensile and Plate Structures.

Course Contents & Hourly break-up

Module and Chapter Name	Course Content	Number of Hours	Number of Tutorials
Module 1: Bending and Shear Stresses in Beams	Theory of pure bending, Theory, General equation of bending,	1	3
	Concept of moment of inertia and sectional modulus,	2	
	Distribution of bending stress in beam cross-section,	5	
	Distribution of shear stress in beam cross-section.	2	
Module 2: Deflection of Beams	Relationship between moment and curvature, deflection of beams by Macaulay's method	2	2
	Deflection of beams: Applications to simply supported, cantilever and overhanging beams.	4	
Module 3: Theory of Columns	Buckling and stability of columns, Euler's theory of column for different support conditions	2	1
	Effective lengths, slenderness ratio,	2	
	Rankine's formula.	2	
Module 4: Analysis of Indeterminate Structures	Statically indeterminacy and degree of indeterminacy, static and kinematic indeterminacy, Degrees of freedom, General methods of analysis of statically indeterminate structures	2	3
	Force method: Analysis by deflection superposition; analysis by three-moment equations: application to analysis of beams, continuous beams.	5	
	Displacement method: Concept of member stiffness and distribution factor, Slope deflection and Moment-distribution method; Analysis of continuous beams and portal frames with and without side sway.	5	
Module 5: Special Structures	Arch, Shell, Dome Tensile and Plate Structures, Structural concept and basic principles,	2	1
	Types, Application and advantages, Case study in ancient and modern architecture.	3	

Total Number of Lecturers: 39
Total Number of Tutorials: 10

Text Book

1. S. P. Timoshenko, and D.H. Young, *Elementa of Strength of Materials*, Affiliated East-West Press.
2. Punmia, B.C., *Structural Analysis*, Standard Publishers Distributors, New Delhi.
3. V. Ramasamy and A. Leena Rose, *Structural Analysis – I & II*, Laxmi Publication
4. Ashok Kumar Jain, *Elementary Structural Analysis*, Nem Chand and Bros. Publication.

Reference books:

1. Negi, L.S., *Theory and Problems in Structural Analysis*, Tata McGraw Hill Publishing Co. Ltd., NewDelhi.
2. Bari, S.A., *Elements of Structural Analysis*, S.Chand & Company Ltd.,NewDelhi.
3. S. Ramamurtham, *Strength of Materials*, Dhanpat Rai Publications
4. S. Ramamurtham and R. Narayan, *Theory of Structures*, Dhanpat Rai Publications
5. S.S. Bhavikatti, *Structural Analysis – I & II*, Vikask Publishing House Pvt. Ltd.
6. S. S. Rattan, *Strength of Materials*, Tata McGraw-Hill Publishers.